

Ushtrime ND7a

1.2.94 Të llogaritet masa e Tokës duke u bazuar në ligjin e gravitetit?

Zgjidhja



Të dhënat:

$$M_T = ?$$

$$F = \gamma \frac{m M_T}{R_T^2}$$

$$F = m g^\oplus = Q$$

$$m g^\oplus = \gamma \frac{m M_T}{R_T^2}$$

$$g^\oplus \cdot R_T^2 = \gamma \cdot M_T / : \gamma$$

$$M_T = \frac{g^\oplus R_T^2}{\gamma}$$

$$R_T = 6380 \text{ km}$$

$$\gamma = 6.67 \cdot 10^{-11} \frac{\text{N} \cdot \text{m}^2}{\text{kg}^2}$$

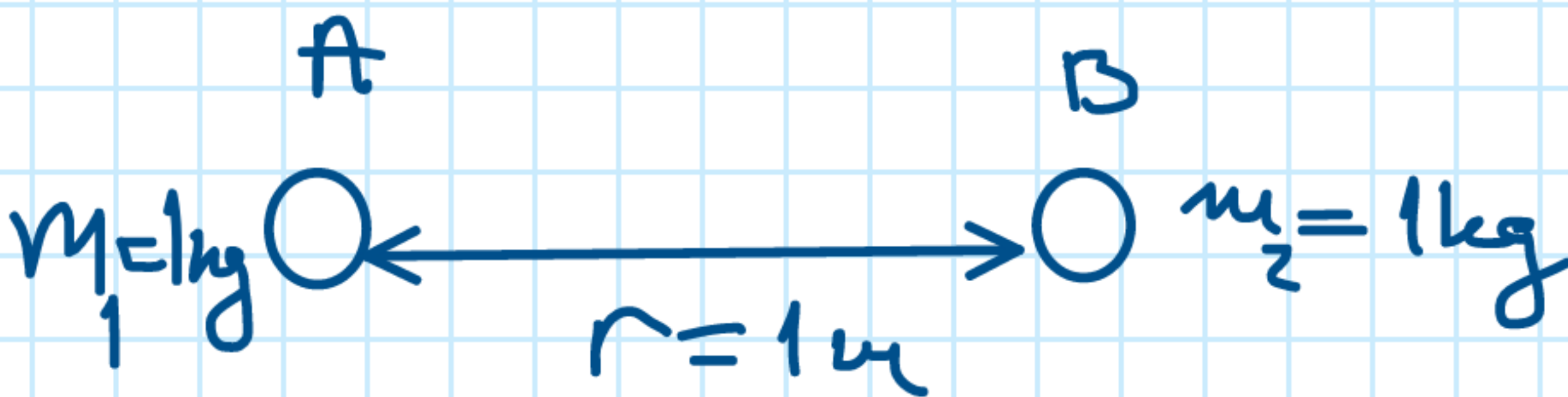
$$g^\oplus = 9.81 \frac{\text{m}}{\text{s}^2}$$

$$M_T = \frac{9.81 \frac{\text{m}}{\text{s}^2} \cdot (6.38 \cdot 10^6 \text{ m})^2}{6.67 \cdot 10^{-11} \frac{\text{N} \cdot \text{m}^2}{\text{kg}^2}} = \frac{9.81 \frac{\text{m}}{\text{s}^2} \cdot 40.7 \cdot 10^{12} \text{ m}^2}{6.67 \cdot 10^{-11} \frac{\text{N} \cdot \text{m}^2}{\text{kg}^2}}$$

$$\frac{\frac{\text{m}}{\text{s}^2} \cdot \text{m}^2}{\frac{\text{N} \cdot \text{m}^2}{\text{kg}^2}} = \frac{\text{m} \cdot \text{kg}^2}{\frac{\text{kg} \cdot \text{m}}{\text{s}^2} \cdot \text{m}^2} = \text{kg}$$

$$M_T = \frac{399,267 \cdot 10^{12}}{6.67 \cdot 10^{-11}} \text{ kg} = 59,86 \cdot 10^{12} \cdot 10^{11} \text{ kg}$$

$$M_T = 5,986 \cdot 10^{24} \text{ kg}$$



RILIND Shabani

$$F = \gamma \frac{m_1 m_2}{r^2}$$

$$F = 6.67 \cdot 10^{-11} \frac{\text{N} \cdot \text{m}^2}{\text{kg}^2} \cdot \frac{1 \text{ kg}^2}{1 \text{ m}^2}$$

$$F = 6,67 \cdot 10^{-11} \text{ N}$$

$$\gamma = 6.67 \cdot 10^{-11} \frac{\text{N} \cdot \text{m}^2}{\text{kg}^2}$$

$$M_T = 5.987 \cdot 10^{24} \text{ kg}$$

$$R_T = 6.38 \cdot 10^6 \text{ m}$$

$$g = ?$$

$$m g = \gamma \frac{m \cdot M_T}{R_T^2}$$

$$g = \gamma \frac{M_T}{R_T^2} = 9,810558 \frac{\text{m}}{\text{s}^2}$$